

MATH 301: Homework 2

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Question 1: Section 2.1-1

If $a, b \in \mathbb{R}$, prove the following:

Part (a): *If $a + b = 0$, then $b = -a$*

Recall Properties of $\mathbb{R}$ Addition:

(Commutativity:) $a + b = b + a$

(Associativity:) $(a + b) + c = a + (b + c)$

(Zero Property:) There exists a $0 \in \mathbb{R}$ such that $a + 0 = a$

(Existence of inverse:) $\forall a \in \mathbb{R}, \exists -a \in \mathbb{R}$ such that $a + (-a) = 0$

Let $a + b = 0$. Observe the following:

$\exists a, b \in \mathbb{R}, b \neq -a \ni$	By proposition
$a + b = 0$	By proposition
$a + b + (-a) = 0 + (-a)$	By existence of inverses
$b + (a + (-a)) = 0 + (-a)$	By associativity
$b + (0) = 0 + (-a)$	By inverse property
$b = -a$	By definition of identity element

Clearly we have shown given $a + b = 0$, then $b = -a$ holds.

Part (c): $(-1)a = -a$

Recall Properties of $\mathbb{R}$ Multiplication

(Commutativity:) $a \cdot b = b \cdot a$

(Associativity:) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

(Identity:) $\exists 1 \in \mathbb{R}$ such that $a \cdot 1 = a$

(Existence of inverse:) $\forall a \neq 0, \exists \frac{1}{a} \in \mathbb{R}$ such that $a \cdot \frac{1}{a} = 1$

Consider the following:

$a + (-1)a = a + (-1)a$	Proposition $x=x$
$a + (-1)a = 1 \cdot a + (-1)a$	Multiplicative Identity
$a + (-1)a = a(1 + (-1))$	Distributive Property
$a + (-1)a = a(0)$	Inverse Property of Addition
$a + (-1)a = 0$	Zero Multiplicative Property

Thus we can say $a + (-1)a = 0$.

Recall by *Part (a)* "If $a + b = 0$, then $b = -a$ ".

Therefore $(-1)a = -a$ must hold.

Part (d): $-(-a) = a$

Consider the following:

$(-a) + a = 0$	Definition of Additive Inverse
$(-1)((-a) + a) = (-1)0$	Multiply both sides by (-1)
$-(-a) + (-a) = (-1)0$	Distributive Property and Part (b)
$-(-a) + (-a) = 0$	Multiplicative Property of Zero
$-(-a) + (-a) + a = 0 + a$	Add a to both sides
$-(-a) + ((-a) + a) = a$	Commutative Property
$-(-a) + (0) = a$	Inverse Property of Addition
$-(-a) = a$	Conclusion

Thus we have shown that $-(-a) = a$.

Part (e): $(-1)(-1) = 1$

Consider $-(-a) = a$, we stated $a \in \mathbb{R}$ as an arbitrary element.

$-(-a) = a$	Part (d)
$-(-1(a)) = 1(a)$	Identity Property of Multiplication
$(-1)(-1(a)) = 1(a)$	Part (c)
$(-1)(-1(1)) = 1(1)$	Let $a = 1$, a is an arbitrary element of $\mathbb{R}$
$(-1)(-1) = 1$	Identity Property of Multiplication

Thus we have shown $(-1)(-1) = 1$

Question 2: Section 2.1-2

If $a, b \in \mathbb{R}$, prove the following:

Part (a): $-(a + b) = (-a) + (-b)$

Consider the following:

$-(a + b) = (-1)(a + b)$	By q1.c
$-(a + b) = (-1)a + (-1)b$	By Distributive Property
$-(a + b) = -a + -b$	By q1.c

Part (b): $(-a) \cdot (-b) = a \cdot b$

Consider the following:

$(-a) \cdot (-b) = (-a) \cdot (-b)$	Proposition $x=-x$
$(-a) \cdot (-b) = (-1) \cdot a \cdot (-1) \cdot b$	By q1.c
$(-a) \cdot (-b) = (-1) \cdot (-1) \cdot a \cdot b$	By Associativity
$(-a) \cdot (-b) = (1) \cdot a \cdot b$	By q1.e
$(-a) \cdot (-b) = a \cdot b$	By Identity Property of Multiplication

Part (c): $\frac{1}{(-a)} = -(\frac{1}{a})$

Consider the following:

$$\begin{array}{ll}
 (\frac{1}{-a}) = -(\frac{1}{a}) & \text{By Proposition} \\
 (\frac{1}{-a}) = (-1)(\frac{1}{a}) & \text{By q1.c} \\
 (\frac{1}{-a})a = (-1)(\frac{1}{a})a & \text{Multiply both sides by } b \\
 (\frac{1}{-a})a = (-1)1 & \text{Property of Multiplicative Inverse} \\
 (\frac{1}{-a})a = (-1) & \text{Property of Multiplicative identity} \\
 (\frac{1}{-a})(-1)a = (-1)(-1) & \text{Multiply both sides by } (-1) \\
 (\frac{1}{-a})(-a) = (-1)(-1) & \text{By q1.c} \\
 1 = (-1)(-1) & \text{Property of Multiplicative Inverse} \\
 1 = 1 & \text{By q1.e}
 \end{array}$$

Question 3: Section 2.1-10

Part (a): If $a < b$ and $c \leq d$, prove that $(a + c) < (b + d)$

Suppose $a < b$ and $c \leq d$. Consider the following:

Recall: (*Theorem*) *Rules of Inequality*

Let $a, b, c \in \mathbb{R}$. The following statements must hold:

- $(a > b) \wedge (b > c) \implies a > c$
- $(a > b) \wedge (c > 0) \implies ca > cb$
- $(a > b) \implies (a + c) > (b + c)$
- $(a > b) \wedge (c < 0) \implies ca < cb$

Applying the theorem above, consider the following:

$$\begin{array}{l}
 a < b \\
 a + c < b + c \\
 b + c \leq b + d \\
 a + c < b + c \leq b + d \\
 a + c < b + d
 \end{array}$$

Part (b): If $0 < a < b$ and $0 \leq c \leq d$, prove that $0 \leq ac \leq bd$

Suppose $0 < a < b$ and $0 \leq c \leq d$

Recall: *(Theorem) Rules of Inequality*

Let $a, b, c \in \mathbb{R}$. The following statements must hold:

- $(a > b) \wedge (b > c) \implies a > c$
- $(a > b) \wedge (c > 0) \implies ca > cb$
- $(a > b) \implies (a + c) > (b + c)$
- $(a > b) \wedge (c < 0) \implies ca < cb$

Ordering If $a - b \in P$, we write $a > b$ (or $b < a$).

If $a - b \in P \cup \{0\}$, we write $a \geq b$ (or $b \leq a$).

For any two real numbers a, b exactly one of the following holds:

$$a > b, a < b, a = b$$

Recall: *Properties of Positive Sets*

Let $P \neq \emptyset, P \subseteq \mathbb{R}$ be called the set of positive real numbers.

1. If $a, b \in P$ then $a + b \in P$.
2. If $a, b \in P$ then $a \cdot b \in P$.
3. *Trichotomy Property:* If $a \in \mathbb{R}$, then exactly one of the following must be true:

$$a \in P, \quad a = 0, \quad -a \in P$$

Applying the theorems above, observe the following:

$$(0 < a) \wedge (0 \leq c) \implies (0 \leq ac)$$

Thus the following statements hold:

$$\begin{aligned} 0 &\leq c \leq d \\ 0 &\leq bc \leq bd \\ 0 &< a < b \\ 0 &\leq ac \leq bc \leq db \\ 0 &\leq ac \leq db \end{aligned}$$

Question 4: Section 2.1-11

Part (a): If $(a > 0)$, then $(1/a > 0)$ and $\frac{1}{1/a} = a$

Suppose $a > 0$ and $\frac{1}{a} \not> 0$. Consider the following:

$$\begin{aligned} \frac{1}{a} &\not> 0 \\ \left(\frac{1}{a}\right) \cdot a &\not> 0 \cdot a \\ \left(\frac{1}{a}\right) \cdot a &\not> 0 \\ 1 &\not> 0 \end{aligned}$$

Clearly $1 > 0$ thus this is a contradiction.

Thus we must accept the original proposition: $a > 0 \implies \frac{1}{a} > 0$.

Prove the following statements $\frac{1}{1/a} = a$:

$$\begin{aligned}\frac{1}{1/a} &= a \\ \left(\frac{1}{1/a}\right)\left(\frac{1}{a}\right) &= a\left(\frac{1}{a}\right) \\ 1 &= 1\end{aligned}$$

Clearly $1 = 1$ thus proven directly.

Now we have proven if $(a > 0)$, then $(1/a > 0)$ and $\frac{1}{1/a} = a$.

Part (b): If $(a < b)$, then $a < \frac{1}{2}(a + b) < b$

Consider the following:

$$\begin{aligned}a &= a \\ \frac{1}{2}a &= \frac{1}{2}a \\ \frac{1}{2}a + \frac{1}{2}a &= \frac{1}{2}a + \frac{1}{2}a \\ \frac{1}{2}(a + a) &= \frac{1}{2}(a + a) \\ \frac{1}{2}(2a) &= \frac{1}{2}(a + a) \\ \frac{1}{2}2(a) &= \frac{1}{2}(a + a) \\ a &= \frac{1}{2}(a + a) \\ a &< \frac{1}{2}(a + b) < b\end{aligned}$$

Question 5: Sec 2.1-13

If $a, b \in \mathbb{R}$, show that $a^2 + b^2 = 0$ iff $a = 0$ and $b = 0$.

Case 1: $\rightarrow$

Prove: If $a, b \in \mathbb{R}$, show that $a^2 + b^2 = 0$ implies $a = 0$ and $b = 0$.

Consider the contrapositive statement, $a \neq 0$ or $b \neq 0$ implies $a^2 + b^2 \neq 0$:

Observe the following:

If $a \in \mathbb{P}$

$$a \in \mathbb{P} \implies a^2 = a \cdot a \in \mathbb{P} \implies a^2 > 0$$

If $-a \in \mathbb{P}$: (Consider $\forall n \in \mathbb{R}, (n - n) \cdot (-n) = 0$)

$$\begin{aligned}n \cdot (-n) + (-n)(-n) &= 0 \\ (-n) + (-n)(-n) &= 0 \\ \implies (-n)(-n) &= n\end{aligned}$$

Therefore, $-a^2 \in \mathbb{P}$. The same logic can be applied to b . Observe If $a, b \in \mathbb{P} \implies a + b \in \mathbb{P}$. Therefore we have proven the contrapositive statement if $a \neq 0$ or $b \neq 0$ implies $a^2 + b^2 \neq 0$

Case 2: $\leftarrow$

Prove: If $a, b \in \mathbb{R}$, show that $a = 0$ and $b = 0$ implies $a^2 + b^2 = 0$.
Consider the following:

$$0 \cdot 0 = 0 \implies \text{If } x = 0, \text{ then } x^2 = x \cdot x = 0 \cdot 0 = 0$$

As $a = b = 0$ then naturally if $a = 0$ and $b = 0$ implies $a^2 + b^2 = 0$

Question 6: Section 2.1-14

If $0 \leq a < b$, show that $a^2 \leq ab < b^2$. Show by example that it does **not** follow that $a^2 < ab < b^2$.
Let $a = 0, b = 1$. Thus $0 \leq 0 < 1$ satisfy $a^2 \leq ab < b^2$. Observe $a^2 < ab < b^2$ does not hold as $0 < 0 < 1$ is false as $0 \not< 0$

Question 7: Section 2.1-20

Part a: If $0 < c < 1$, show that $0 < c^2 < c < 1$.

We can define $1 < c$ as $1 < 1 - \epsilon$ where $c = 1 - \epsilon = 1 - \frac{1}{n}, n \in \mathbb{N}$.
Observe the following:

$$\begin{aligned} 1 &< (1 - \epsilon) \\ 1^2 &< (1 - \epsilon)^2 \\ 1^2 &< \left(1 - \frac{1}{n}\right)^2 \\ \left(1 - \frac{1}{n}\right)^2 &= 1 - \frac{2}{n} + \frac{1}{n^2} \\ \text{Observe } \frac{1}{n} &> \frac{1}{n^2} \implies \end{aligned}$$

$$1 > \left(1 - \frac{1}{n}\right) > \left(1 - \frac{2}{n} + \frac{1}{n^2}\right)$$

Thus we can conclude: $1 > 1 - \epsilon > (1 - \epsilon)^2$

Part b: If $1 < c$, show that $1 < c < c^2$.

We can define $1 < c$ as $1 < 1 + \epsilon$ where $c = 1 + \epsilon = 1 + \frac{1}{n}, n \in \mathbb{N}$.
Observe the following:

$$\begin{aligned} 1 &< (1 + \epsilon) \\ 1^2 &< (1 + \epsilon)^2 \\ (1 + \epsilon)^2 &= 1 + 2\epsilon + 1 \\ 1 &< 1 + \epsilon < 1 + 2\epsilon + 1 \\ 1 &< c < c^2 \end{aligned}$$

As we defined $\epsilon > 0$.

The statement $1 < c < c^2$ holds

Question 8: Section 2.1-23

If $a > 0$, $b > 0$, and $n \in \mathbb{N}$, show that $a < b$ iff $a^n < b^n$ (induct.)

$\forall a, b \in \mathbb{R}, a > 0, b > 0$ and $n \in \mathbb{N}$.

Consider $P(n) : a^n < b^n \iff a < b$

Observe $P(1)$:

$$a^1 < b^1 \implies a < b$$

$$a < b \implies a^1 < b^1$$

Suppose $P(k), k \in \mathbb{N}$:

$$a^k < b^k \implies a < b$$

Observe the following:

$$a^k < b^k \implies a < b$$

$$a^k < b^k$$

$$a^k \cdot ab < b^k \cdot ab$$

$$a^{k+1}b < b^{k+1}a$$